Tarea sección 3.6

3) $f(x) = \operatorname{sen}(\ln x)$

$$f'(x) = \cos(\ln x) * \frac{1}{x} = \frac{\cos(\ln x)}{x}$$

9) $g(x) = \ln(x e^{-2x})$

$$g(x) = \frac{1}{x e^{-2x}} * e^{-2x} + x e^{-2x} * -2$$

$$g(x) = \frac{1}{x} - 2$$

13) $g(x) = \ln(x\sqrt{x^2-1})$

$$g'(x) = \frac{1}{x\sqrt{x^2-1}} * \left(\sqrt{x^2-1} + x * \frac{1}{2\sqrt{x^2-1}} * 2x\right)$$

$$g'(x) = \frac{2x^2-1}{x^3-x}$$

19) $y = \ln(e^{-x} + xe^{-x})$

$$y' = \frac{1}{e^{-x}+xe^{-x}} * -e^{-x} + x(e^{-x}(-x)) + e^{-x}(x)$$

$$y' = \frac{xe^{-x}}{e^{-x}+xe^{-x}} = -\frac{x}{1+x}$$

$y = \tan[\ln(ax+b)]$ 220

$y' = \sec^2[\ln(ax+b)] * \frac{1}{ax+b} * a$

$y' = \frac{\sec^2[\ln(ax+b)]a}{ax+b} = \frac{a\sec^2[\ln(ax+b)]}{ax+b}$

23) $y = \sqrt{x}\ln x$

$y' = \frac{1}{x} * \ln x + \sqrt{x} * \frac{1}{x}$

$y' = \frac{\ln x}{2\sqrt{x}} + \frac{1}{\sqrt{x}} = \frac{\ln x + 2}{2\sqrt{x}}$

$y'' = \frac{\left(\frac{1}{x} * 2\sqrt{x}\right) - \ln x + \frac{2}{2x}}{(2\sqrt{x})^2}$

$y'' = \frac{\frac{2}{x} - \frac{\ln x + 2}{\sqrt{x}}}{4x}$

$y'' = \frac{\frac{2 - \ln x - 2}{\sqrt{x}}}{\frac{4x}{1}} = \frac{\ln x}{4x\sqrt{x}}$

25) $y = \ln|\sec x|$

$y' = \frac{1}{|\sec x|} * \frac{\sec x}{|\sec x|}(\sec x)$

$y' = \frac{\sec x \sec x}{|\sec^2 x|}\tan x = \frac{\sec^2 x \tan x}{\sec^2 x} = \tan x$

$y'' = \tan x = \sec^2 x$

27) $F(x) = \frac{x}{1 - \ln(x-1)} =$

$f'(x) = \frac{[1 - \ln(x-1)(1) - x[-\frac{1}{x-1}]]}{[1 - \ln(x-1)]^2}$

$f'(x) = \frac{1 - \ln(x-1) + \frac{x}{x-1}}{[1 - \ln(x-1)]^2}$

$\ln(x-1) * x - 1 > 0$

$x > 1$

$f'(x) = \frac{x - 1 - (x-1)\ln(x-1) + x}{(x-1)[1 - \ln(x-1)]^2}$

$D = (1, 3.72) \cup (3.72, \infty)$

31) Si $f(x) = \ln(x+\ln x)$, determine $f'(1)$

$$f'(x) = \frac{1}{x+\ln x} * 1 + \frac{1}{x} = \frac{x+1}{x(x+\ln x)}$$

$$f'(1) = \frac{1+1}{1(1+\ln 1)} = \frac{2}{1} = 2$$

33) $y = \ln(x^2-3x+1)$ $\quad (3,0)$

$$y' = \frac{1}{x^2-3x+1} * 2x - 3 = \frac{2x-3}{x^2-3x+1}$$

$$y'|_3 = \frac{2(3)-3}{3^2-3(3)+1} = \frac{3}{1} = 3$$

$$y-0 = 3(x-3)$$
$$y = 3x-9$$

37) dado $f(x) = cx + \ln(\cos x)$. ¿Para qué valores de c es $f'(\pi/4) = 6$?

$$f'(x) = c + \frac{1}{\cos x} \cdot \operatorname{sen} x$$

$$f'(x) = c - \frac{\operatorname{sen} x}{\cos x}$$

$$f'(\pi/4) = c - \tan \pi/4$$
$$f(\pi/4) = c = 1$$

c	$f(\pi/4)$
2	1
1	0
0	-1
-1	-2
-2	-3
-5	-6

c	$(\pi/4)$
10	9
9	8
8	7
7	6
4	3
3	2

41) $y = \sqrt{\frac{x-1}{x^4+1}}$

$$\ln y = 1/2\left(\ln(x-1) - (x^4+)\right)$$

$$y'/y = 1/2\left(\ln(x-1) - \ln(x^4+1)\right)$$

$$\frac{y'}{y} = -\frac{3x^4-4x^3-1}{2(x-1)(x^4+1)} = y' = -\frac{3x^4-4x^3-1}{2(x-1)(x^4+1)}\sqrt{\frac{x-1}{x^4+1}}$$

4a) $y = (\tan x)^{1/x}$

$$\ln y = \frac{\ln(\tan x)}{x} = \frac{y'}{y} = \frac{x \csc x \sec x - \ln(\tan x)}{x^2} =$$

$$\frac{x \cos x \sec x - \ln(\tan x)}{x^2} = -\frac{\ln(\tan x)}{x^2}$$